

Personalization Increases Affective Alignment but Has Role-Dependent Effects on Epistemic Independence in LLMs

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Abstract

Large Language Models (LLMs) are prone to sycophantic behavior, uncritically conforming to user beliefs. As models increasingly condition responses on user-specific context (personality traits, preferences, conversation history), they gain information to tailor agreement more effectively. Understanding how personalization modulates sycophancy is critical, yet systematic evaluation across models and contexts remains limited. We present a rigorous evaluation of personalization’s impact on LLM sycophancy across nine frontier models and five benchmark datasets spanning advice, moral judgment, and debate contexts. We find that personalization generally increases affective alignment (emotional validation, hedging/deference), but affects epistemic alignment (belief adoption, position stability, resistance to influence) with context-dependent role modulation. When the LLM’s role is to give advice, personalization strengthens epistemic independence (models challenge user presuppositions). When its role is that of a social peer, personalization decreases epistemic independence. In this role, extensively personalized user challenges causing LLMs to abandon their position at significantly higher rates. Robustness tests confirm that the effects are driven by personalized conditioning, not by additional input tokens per se or demographic information alone. Our work provides measurement frameworks for evaluating personalized AI systems, demonstrates the necessity of role-sensitive evaluation, and establishes a novel benchmark to assess goal alignment.

actions creates new evaluation challenges (Riedl & Weidmann, 2025). In contexts without verifiable ground-truth, how do we assess whether models are helpful or merely agreeable? Understanding this distinction requires frameworks for measuring different dimensions of model behavior and characterizing how they vary across interaction contexts. Sycophancy, where models excessively conform to user beliefs and provide biased validation, has emerged as a key evaluation concern (Sharma et al., 2023). Sycophancy manifests through biased LLM responses, providing answers that conform to user beliefs, and readily agreeing with users when challenged. These behaviors are thought to arise from optimization in post-training via reinforcement learning from human feedback (RLHF) (Perez et al., 2023; Lambert, 2025). Even simple triggers like adding opinion statements (e.g., “I believe...”) to prompts can elicit sycophantic responses (Wang et al., 2025a), and models become progressively more agreeable across multi-turn conversations when subjected to persistent challenges (Hong et al., 2025; Liu et al., 2025a). Recent case studies document instances where excessive validation may reinforce problematic beliefs (Preda, 2025; Archiwaranguprok et al., 2025; Morrin et al., 2025), prompting model developers, like OpenAI, to implement targeted mitigation strategies (OpenAI, 2025).

While early sycophancy research focused on factual errors (e.g., models agreeing with incorrect statements) (Sharma et al., 2023; Hill, 2025; Laban et al., 2023), recent work has expanded to encompass social sycophancy, which involves validation in subjective contexts related to self-image, emotions, and personal decisions (Cheng et al., 2025b). In these open-ended domains lacking ground-truth answers, sycophancy emerges through subtle language patterns. A naturalistic study of ChatGPT found strong tendencies to generate empathetic responses, validate emotions, and encourage self-care, particularly in personal conversations (Fang et al., 2025). Cheng et al. (2025b) demonstrated that current frontier LLMs display significantly higher levels of emotional validation, hedging (deference), and acceptance of user problem framings compared to human responses. Other negative consequences of sycophantic are that it leads users to judge their own behavior as more morally righteous and reduce willingness to repair interpersonal conflicts (Cheng et al., 2025a). Previous attempts to alter models’ speech patterns through fine-tuning can have unforeseeable

1. Introduction

Large Language Models (LLMs) are increasingly used for personal and professional support (Chatterji et al., 2025). This shift toward subjective, open-ended multi-turn inter-

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consequences (Ibrahim et al., 2025). Despite its importance, measuring and evaluating the varied manifestations of sycophantic responses remains challenging. It is unclear how to detect and characterize changes in LLM responses.

A fundamental challenge is distinguishing between the form and content of sycophancy’s influence (Ryan et al., 2025). When models respond warmly and validate users’ emotions, are they adapting communication style to aid understanding, or adjusting truth-values to maximize approval? Building on prior work (Fiske et al., 2007; Clark, 2015; Clark & Brennan, 1991; Deutsch & Gerard, 1955) we propose that evaluation requires measuring two orthogonal dimensions. **Affective alignment** concerns emotional validation, empathy, tone, and rapport. **Epistemic alignment** concerns belief adoption, opinion stability, and resistance to social influence. Ideally, personalization would target only affective dimensions (adapting how models communicate) while preserving epistemic independence (maintaining accuracy regardless of user beliefs; cf. Deutsch & Gerard, 1955). Current systems may fail to disentangle these dimensions, simultaneously adapting both form and content. Without measuring these dimensions separately, we cannot determine whether observed changes represent appropriate personalization or excessive accommodation.

By incorporating user preferences and memory of previous interactions, models can produce responses aligned with user abilities and needs (Zhang et al., 2024), scaffolding conversations and assisting users to produce higher-quality work (Kelley et al., 2025). However, personalization also provides models with user-specific information that could enable different types of alignment. Prior work suggests concerning patterns. LLMs with personality traits like high agreeableness generate content with differential levels of bias and toxicity (Wang et al., 2025b), and users report higher trust in models aligned with their opinions (Eder et al., 2025). The extent to which personalization affects affective versus epistemic dimensions, and whether these effects vary across contexts (roles the LLM plays), is unclear. From a computational perspective, personalization provides features that enable more efficient optimization of user preferences. Models gain information needed to predict which responses maximize reward signals (whether next-token prediction likelihood or RLHF reward), effectively reducing output entropy to match user priors (Wang et al., 2024a). As AI become better at modeling user preferences (Prakash et al., 2025), personalization acts as a hyperparameter increasing weight on predicted patterns, though what patterns are predicted may differ by context (Park et al., 2024; Poddar et al., 2024; Wen et al., 2024).

We hypothesize that dialogue context establishes roles with distinct behavioral priorities (advisor, peer), and personalization amplifies role-consistent behavior by providing

information that models interpret as role-appropriate justifications. Building on Deutsch & Gerard’s framework of informational versus normative social influence (Deutsch & Gerard, 1955), we propose that models reproduce different influence types because training data contains distinct patterns for different roles. In advisor contexts, personalization may activate informational influence patterns by enabling diagnostic justifications that strengthen epistemic independence. In peer contexts, personalization may activate normative influence patterns by increasing relationship salience and prioritizing affective alignment over epistemic commitments, effectively increasing the penalty for disagreement as models accommodate positions to maintain common ground¹. Notably, in factual domains (MMLU-Pro), we find personalization does not systematically amplify sycophancy with answer sycophancy instead driven by the user’s challenge rather than identity.

In this study, we rigorously evaluate how personalization affects LLM behavior across nine frontier models (both open and closed weights) and five benchmark datasets. We classify these contexts into two distinct conversational roles: **advisory** (spanning personal advice and moral judgment) and **peer** (spanning debate and goal alignment), alongside a domain knowledge control. We develop novel approaches to independently quantify affective and epistemic alignment and characterize how they behave in varied circumstances. We find that personalization uniformly increases affective alignment, but epistemic effects depend on context and the LLM’s conversational role. In open-ended personal advice queries (OEQ; Cheng et al., 2025b), personalization increases both affective alignment and epistemic independence. Models challenge users’ problem framings by offering a diagnostic reframing grounded in user traits. In moral judgment contexts (AITA; Cheng et al., 2025b), personalization increases affective alignment without affecting epistemic dimensions. In debate settings (SYCON-Bench; Hong et al., 2025), personalization increases affective alignment while decreasing epistemic independence. Models become more open to user positions and, under sustained multi-turn challenges with extensively personalized rebuttals, abandon their positions at significantly higher rates than with generic challenges.

We make three main contributions:

- **A critical re-evaluation of AI personalization utility:** We identify a role-dependent trade-off where personalization enhances utility in advisory contexts, but compromises independence in peer interactions by increasing susceptibility to opinion drift. Using our open-

¹While other roles are possible, such as teacher, mediator, critic, etc., but focus on these two roles because they directly align with the Deutsch and Gerard model of normative vs. informational influence.

source **GoalPref-Bench**² dataset, we evaluate 9 state-of-the-art models and demonstrate that personalization-induced sycophancy is a prevalent systematic bias in open-ended dialogue.

- **A role-differentiated evaluation framework:** We propose a novel framework for assessing LLM alignment that distinguishes between affective dimensions (emotional validation, hedging) and epistemic dimensions (framing acceptance, goal alignment).
- **Isolation of personalization as the active mechanism for bias:** We demonstrate that frontier models actively modulate their sycophancy based on inferred user preferences, significantly increasing levels “openness” when they predict a user’s latent preference for validation.

2. Methods

Our main analysis combines five independent datasets (Table S1) to evaluate LLMs for sycophantic language and behavior. To provide a broad evaluation across the current landscape of leading LLMs we repeat each analysis using the same response-generation procedure (temperature = 0.7) across nine state-of-the-art systems that include a mix of open and closed weight models with a range of parameter sizes: DEEPSEEK v3.1, QWEN 2.5 72B INSTRUCT, QWEN3 235B A22B 2507 INSTRUCT, LLAMA 3.1 8B INSTRUCT, LLAMA 3.3 70B INSTRUCT, MISTRAL SMALL 3.2 24B, GROK 3 MINI, GEMINI 2.5 FLASH, and CLAUDE HAIKU 4.5.

2.1. Construction of User Personas

Personalization was operationalized as the addition of demographic, personality, and trait-based information (i.e., age, gender, employment status, education level, socioeconomic status, fluid and emotional intelligence, and Big Five personality traits) randomly sampled to create user personas that preceded each prompt. All traits were independently sampled with equal probability from predefined categorical levels (Table S2). Personas were constructed by prompting GPT-4o to “create a concise character persona based on the [following] demographic and personality trait data” (see Appendix A.3.1). This process was then repeated 500 times to generate a sample of personas to draw upon. By randomly sampling from a diverse persona pool, we can attribute any observed sycophantic behavior to the presence of personalization information per se, rather than to idiosyncratic responses to specific trait levels or combinations. Sycophantic challenges are segmented into one of 3 treatment conditions: generic, personalized, and personalized+.

²<https://github.com/seanwkelley/GoalPref-Bench/>

2.2. Social Sycophancy: Open-Ended Questions and Reddit’s r/AmITheAsshole (AITA)

Open-Ended Questions (OEQ) is a 3,027 set of personal advice questions, derived from multiple sub-Reddit forums, to assess LLM responses in cases without ground-truth answers (Cheng et al., 2025b). We randomly selected a subset of 250 questions, processed the questions to remove personal information, and extracted the primary personal question. This process enables us to precisely control the level of personalization when evaluating AI responses and control for differences in length, question framing, syntax, and presentation of backstory details. r/AmITheAsshole (AITA) is a Reddit forum where users can pose questions pertaining to their own personal situation and ask whether they are at fault or otherwise responsible. We selected a random subset of 250 questions (out of 2,000 possible), in which the user was judged to be at fault in the situation by the group consensus, i.e., top comment on post said the user is to blame. Like with the OEQ dataset, we pre-process all questions and extract: i) core questions (“is the user justified in ...”) and ii) key problem, i.e., description of the situation presented with neutral non-leading language (Figure S1).

For the generic treatment condition, we then prompt the LLM with either i) question only (OEQ) or ii) question + key problem (AITA) (see Appendix A.3.2). While in the personalized condition, we append a randomly selected user persona after presenting the relevant question. The LLM is not provided with any instructions on how to use the persona when generating its response. This process produces a generic and personalized AI response to each question. Similar to the ELEPHANT framework (Cheng et al., 2025b), we measure the relative difference in language between treatment conditions for emotional validation, hedging/deference, framing acceptance, and responsibility attribution (AITA only).

2.3. SYCON-Bench: Debate

We build on the debate domain from the SYCON-Bench dataset to assess the effect of opinion challenges to an LLM’s positional stability³(Hong et al., 2025). SYCON-Debate consists of 100 arguments, paired with 100 questions, across a wide variety of topics, e.g., environmental concerns, health, and education. We prompt AI (GPT-4o, temperature = 0.7) to adopt the viewpoint of each argument (e.g., “the production and use of biofuel is a sustainable solution to reducing global carbon emissions”) and ask it to respond to the associated question (e.g., “is the production and use of biofuel a sustainable solution to reducing global carbon emissions?”) (see Appendix A.3.3). This

³SYCON-Bench has two other domains (ethical challenges and false presuppositions), however, because these are less focused on open-ended feedback/discussion we use only the debate domain